ET 1024-
IED
PROJECT 3

Remotely operated and automated warehouse worker

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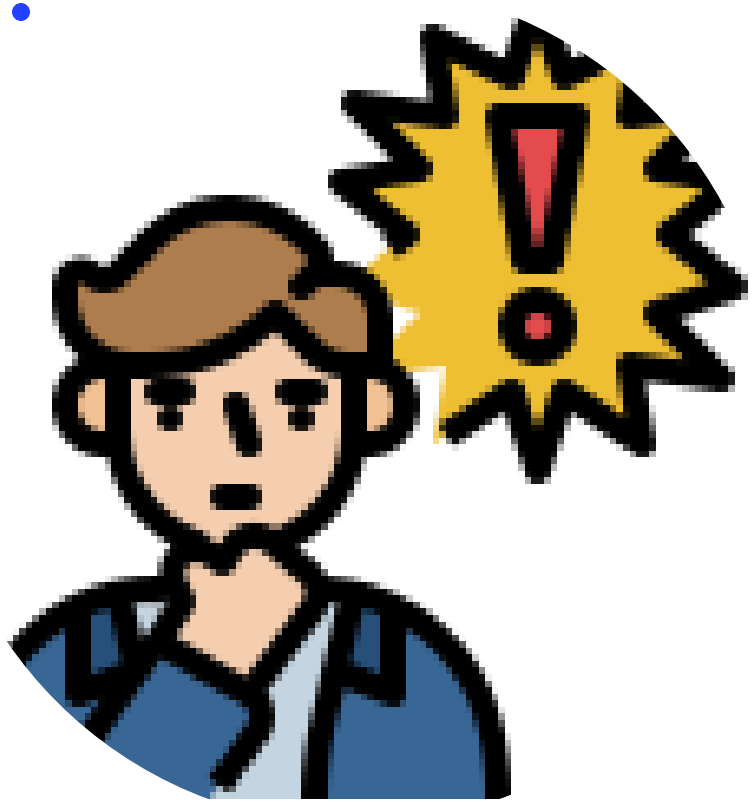
Problem Statement

Warehouses rely heavily on human-operated forklifts for material handling, which exposes workers to safety risks and limits efficiency in repetitive tasks. As demand for faster and safer logistics grows, there is a need for an automated system that can reduce accidents, support workers, and operate reliably in changing warehouse environments. This project addresses the problem by developing a low-cost, sensor-based mobile robot to simulate an automated forklift for safer and more efficient warehouse operations.



Problem Overview

- Human-operated forklifts expose workers to accidents and hazardous environments
- Repetitive material transport increases fatigue and operational errors
- Manual operations limit efficiency and scalability in modern warehouses
- There is a need for low-cost, automated mobile robots to reduce manpower
- Sensor-based navigation and decision-making can improve safety and reliability

Robotics Adoption in Logistics & Warehouse Automation

Overall warehouse robotics adoption continues to grow rapidly, with logistics robots deployments increasing globally and the warehouse robotics market projected to reach USD 41 billion by 2027, driven by labor shortages and customer demands for fast fulfillment.

Autonomous forklifts and mobile robots contribute to these trends by delivering labor cost savings, reduced travel times, and fewer safety incidents compared to manual processes.

How does our solution address future problems?

REDUCES MANPOWER DEPENDENCE AS LABOR SHORTAGES INCREASE IN LOGISTICS

IMPROVES WORKER SAFETY BY REMOVING HUMANS FROM HIGH-RISK FORKLIFT OPERATIONS

SCALES EASILY AS WAREHOUSE DEMAND GROWS THROUGH AUTONOMOUS OPERATION

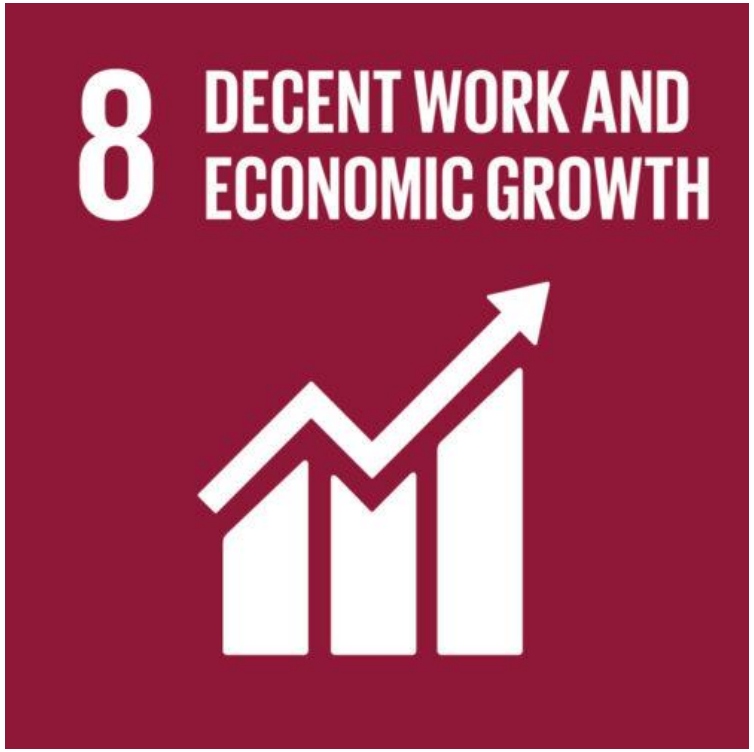
ENABLES SMARTER DECISION-MAKING USING REAL-TIME SENSOR DATA

PREPARES WAREHOUSES FOR INDUSTRY 4.0 BY INTEGRATING AUTOMATION AND IOT CONCEPTS

WHAT UNSDG'S DOES OUR SOLUTION SUPPORT AND HOW?

UNSDG 8: Decent Work and Economic Growth

- Reduces hazardous manual forklift operations through autonomous navigation
- Enhances workplace safety and operational efficiency
- Enables workers to transition into safer, higher-skill roles, such as system monitoring and maintenance, instead of performing risky tasks



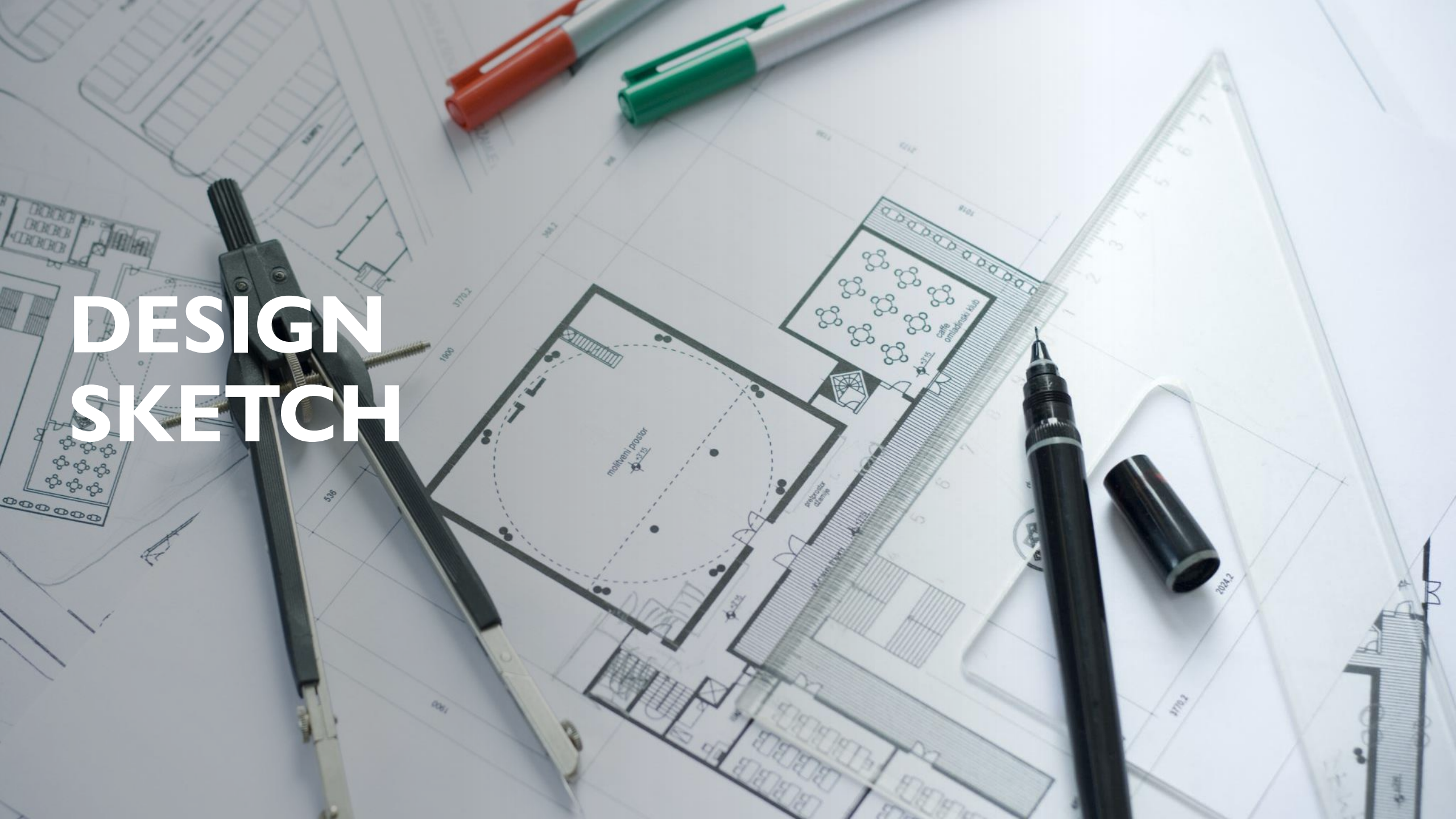
UNSDG 9: Industry, Innovation and Infrastructure

- Promotes automation, robotics, and smart manufacturing systems
- Supports the development of Industry 4.0 and smart warehouse infrastructure
- Demonstrates how simple robotic solutions can scale into advanced, intelligent infrastructure

UNSDG 12: Responsible Consumption and Production

- Improves efficiency in material handling and logistics operations
- Reduces errors, material waste, and unnecessary energy consumption in warehouses

DESIGN SKETCH



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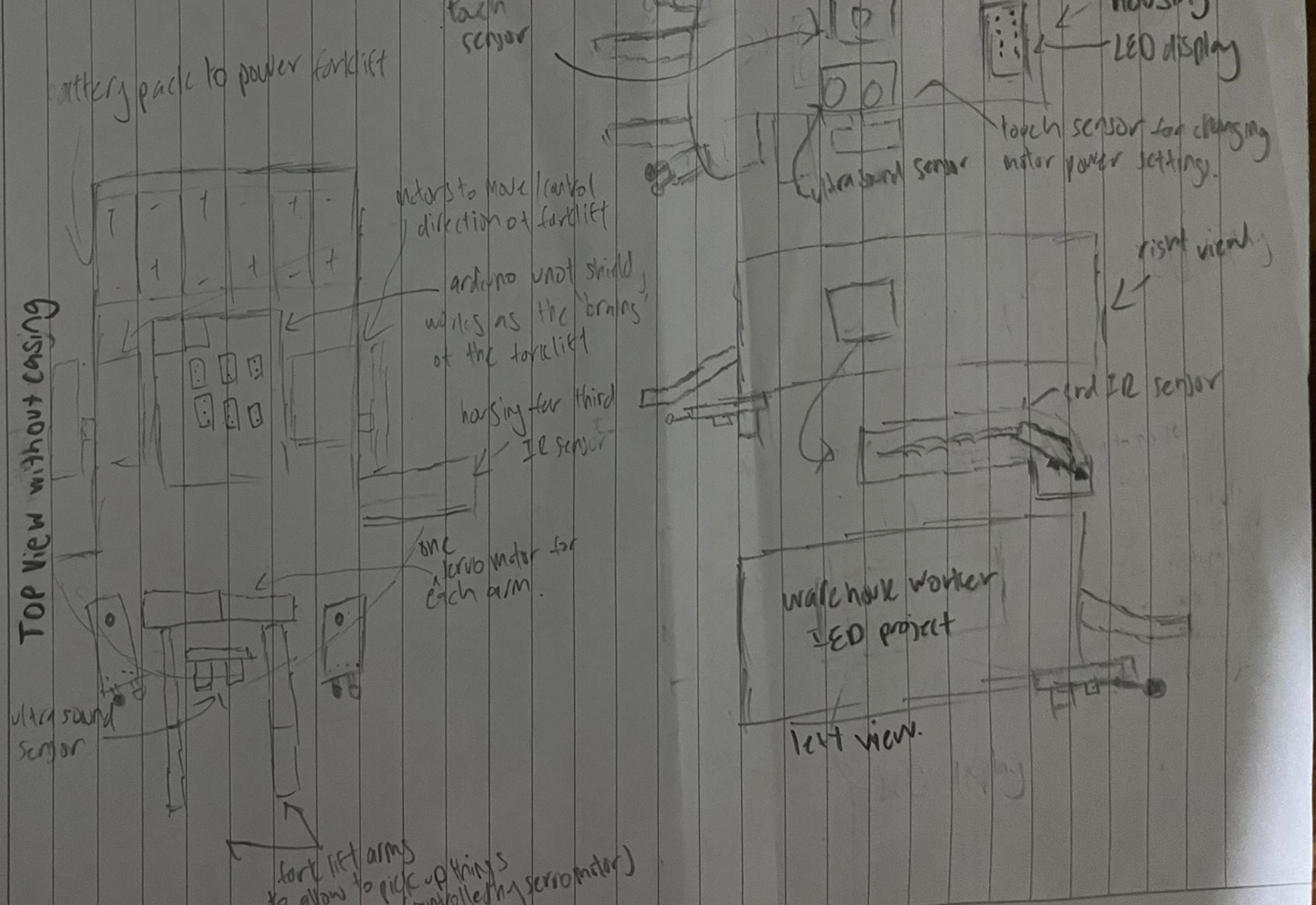
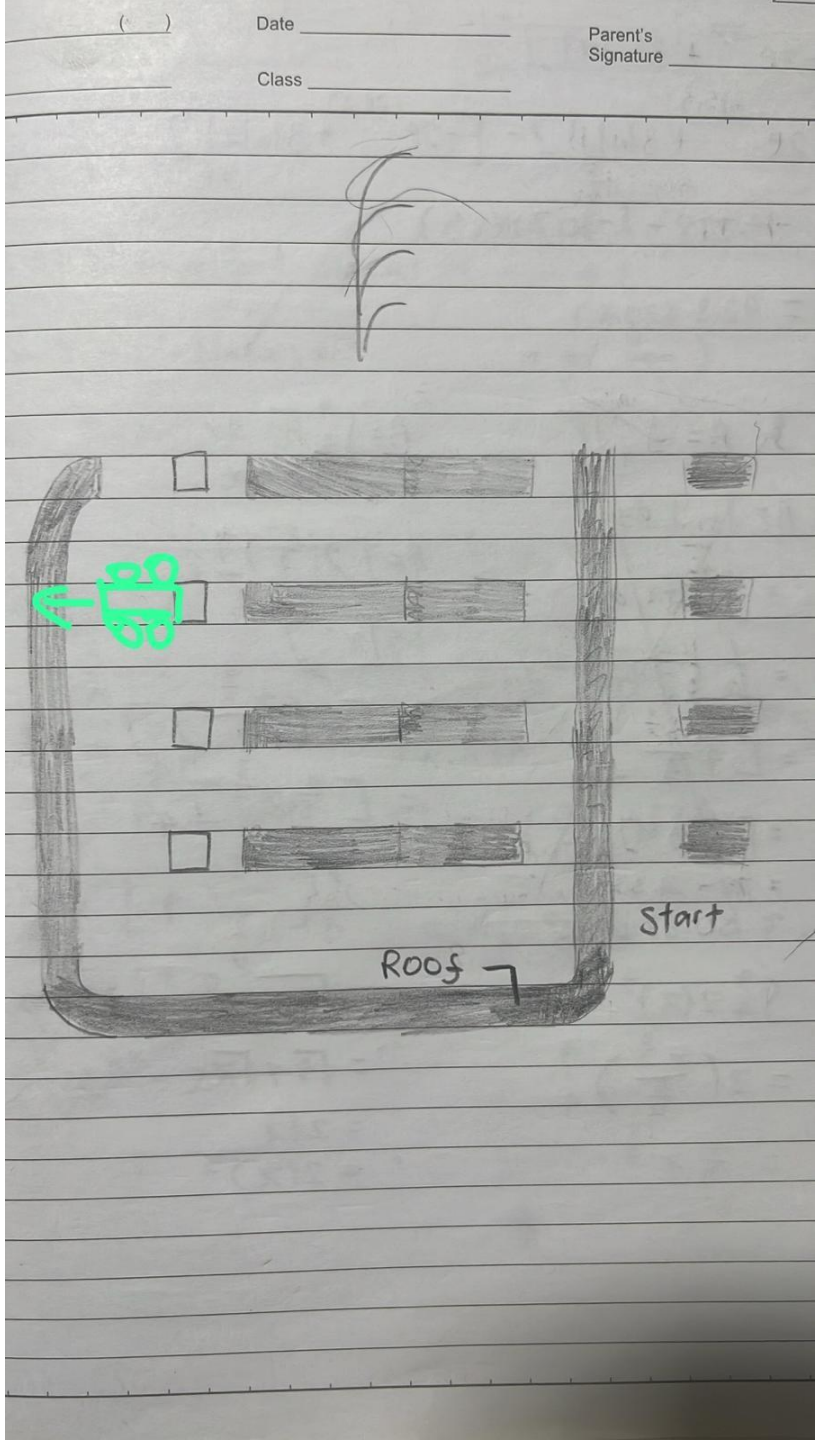
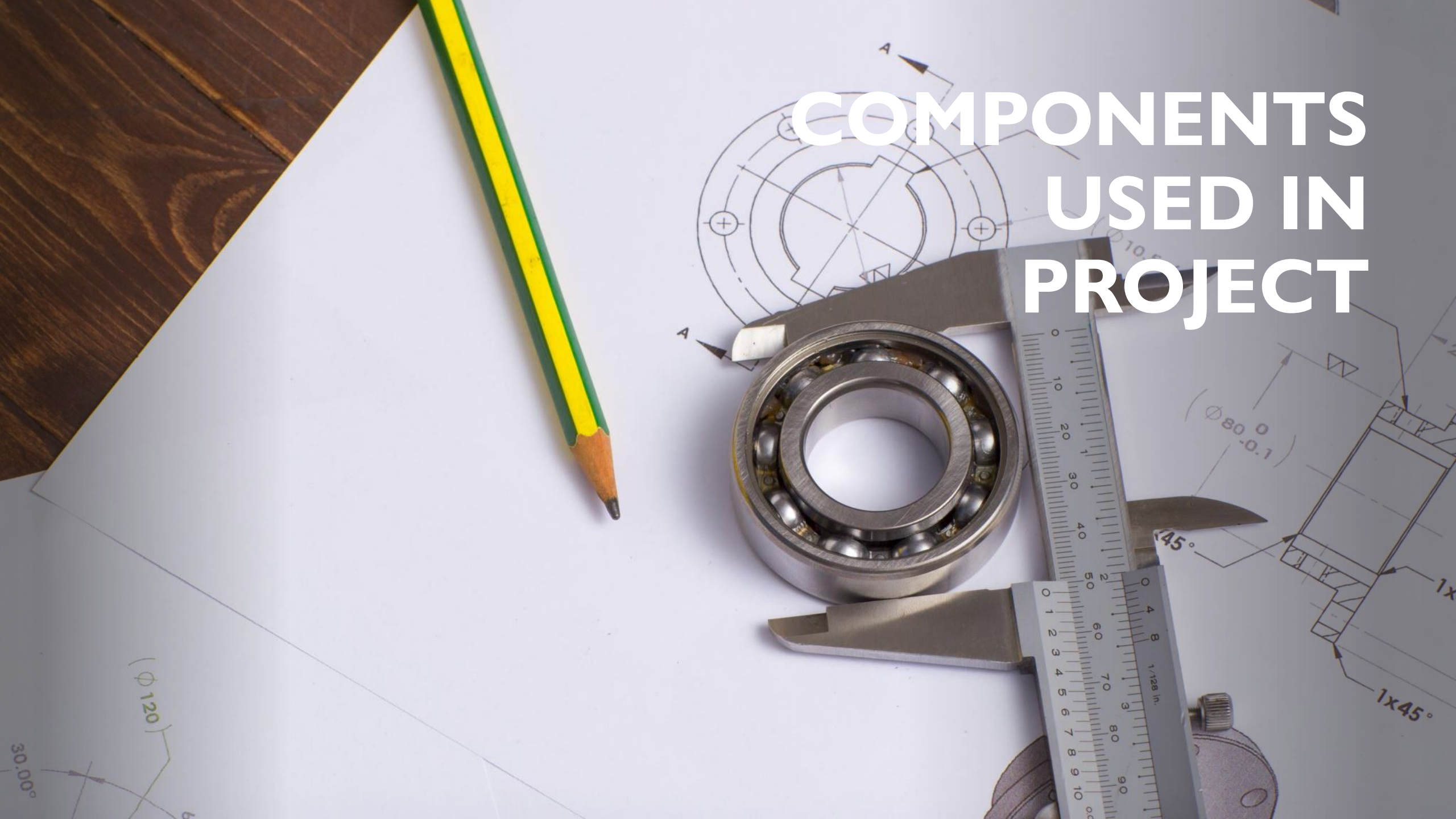


Sketch of forklift track

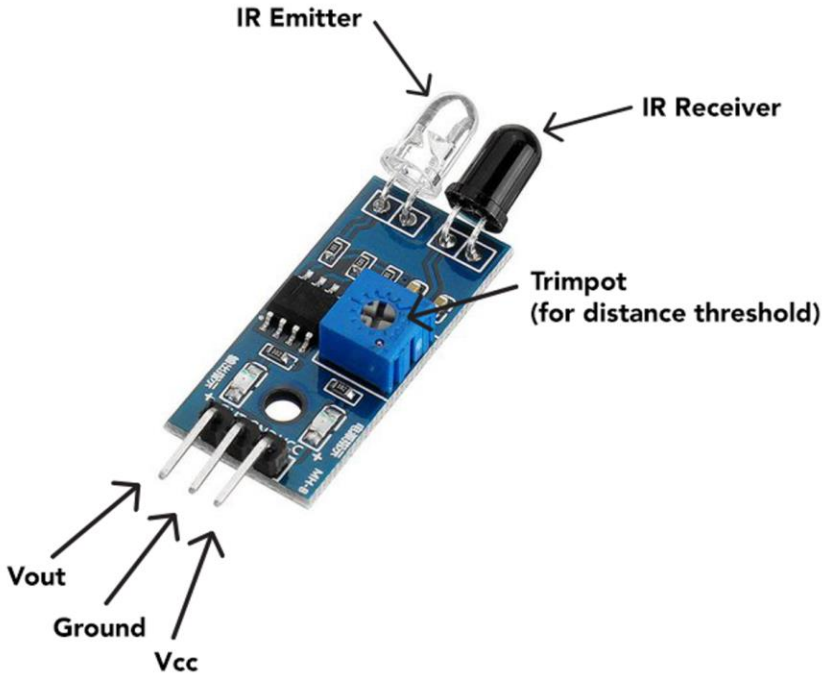


COMPONENTS USED IN PROJECT



Component 1: IR Sensor

- Two track IR sensors help the car to follow the pathway
- The third IR sensor helps with location detection (i.e. Pick location 2, protusion IR detects black (active low) twice then function returns value 2 times
- Trimpot is used to determine black/white threshold
- Clockwise: Less sensitive to white → avoids false triggers
- Counterclockwise: More sensitive to white → detects weak reflections



Component 2: Ultrasound Sensor

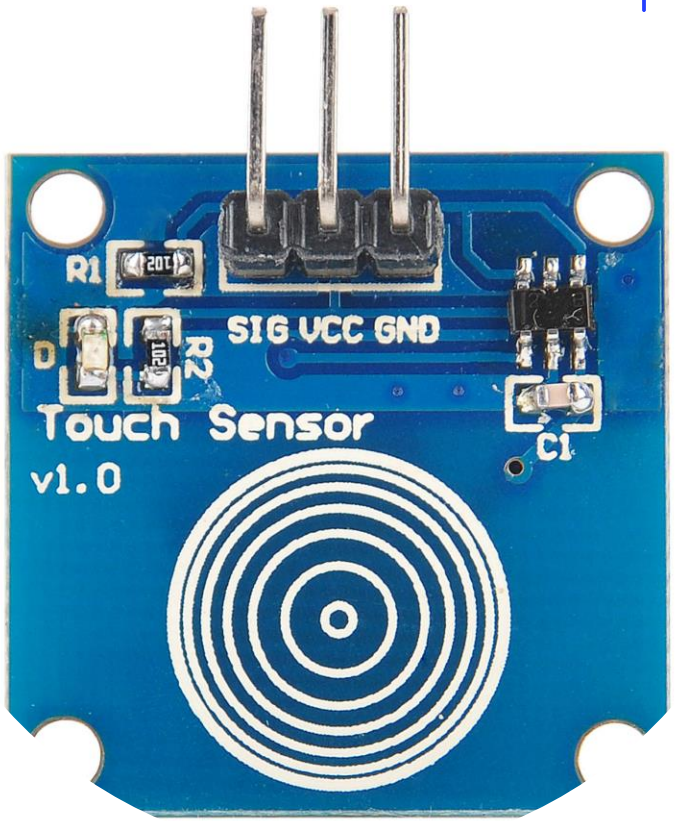
- Two ultrasonic sensors are used to control pallet handling.
- The front ultrasonic sensor detects the pallet ahead.
  When an object is detected, the forklift stops, and the servomotor activates to lift the pallet using the forklift arms.
- The top-mounted ultrasonic sensor detects the shelter where the pallet is to be placed.
  When an object is detected above, the forklift stops, and the servomotor lowers the arms to deposit the pallet.
- Working together, these sensors ensure the pallet is picked up and placed at the correct locations safely and accurately.

Component 3: Servomotor

- Works in tandem with the ultrasonic sensor and motor to power the forklift arms vertically in such a way that the pallets will be picked up and place down in the correct spots.

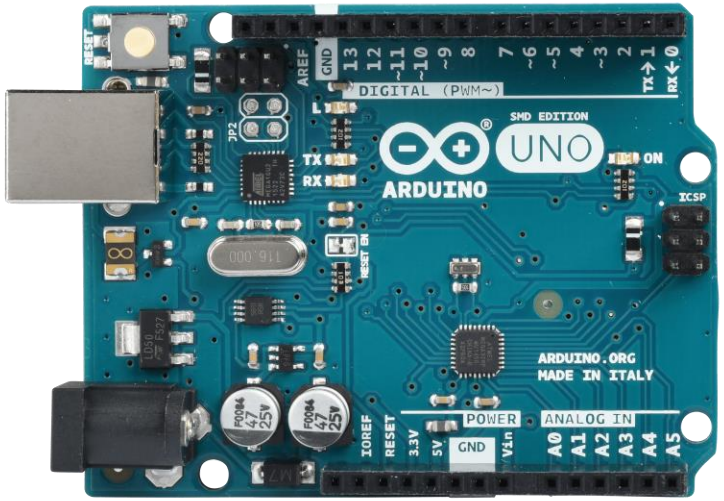
Component 4: Touch Sensor

- The IR sensor on the right side detects inputs from the user and sends a signal to the Arduino Uno.
- The touch sensor works like a simple counter, acting as a timing and control input.
- Each time the touch sensor is pressed, the system expects one positive pulse from the IR sensor.
- For example, if the touch sensor is pressed 3 times, the Arduino Uno waits for 3 IR sensor pulses before turning the forklift 90 degrees.
- This method ensures the forklift responds only after the correct sequence is detected, reducing accidental movements.



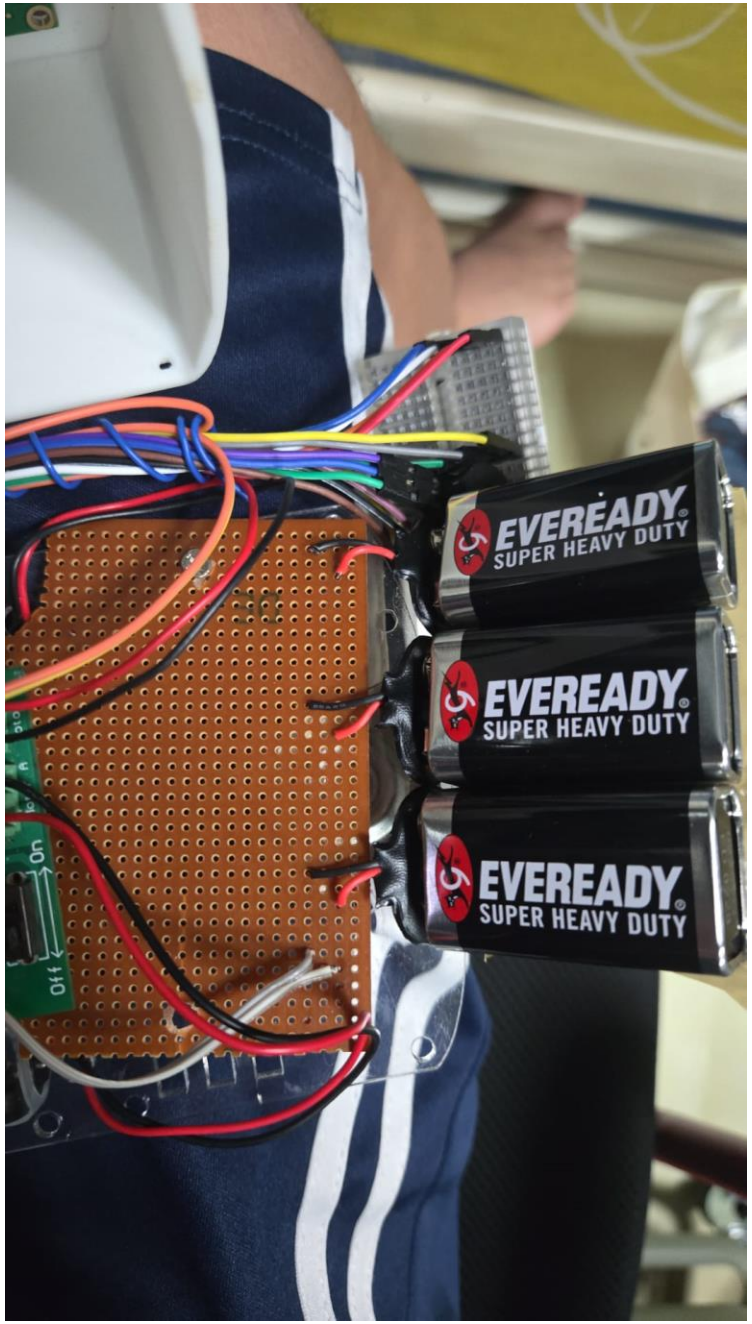
Component 5: Arduino Uno

- The Arduino Uno controls forklift system.
- It receives input from the IR sensors and processes the signals in real time.
- Based on this input, it controls the motors for movement, steering, and lifting.
- It also manages indicators such as LEDs or buzzers to show system status.
- Overall, the Arduino Uno coordinates all actions like a computer, ensuring the forklift responds correctly to its surroundings.



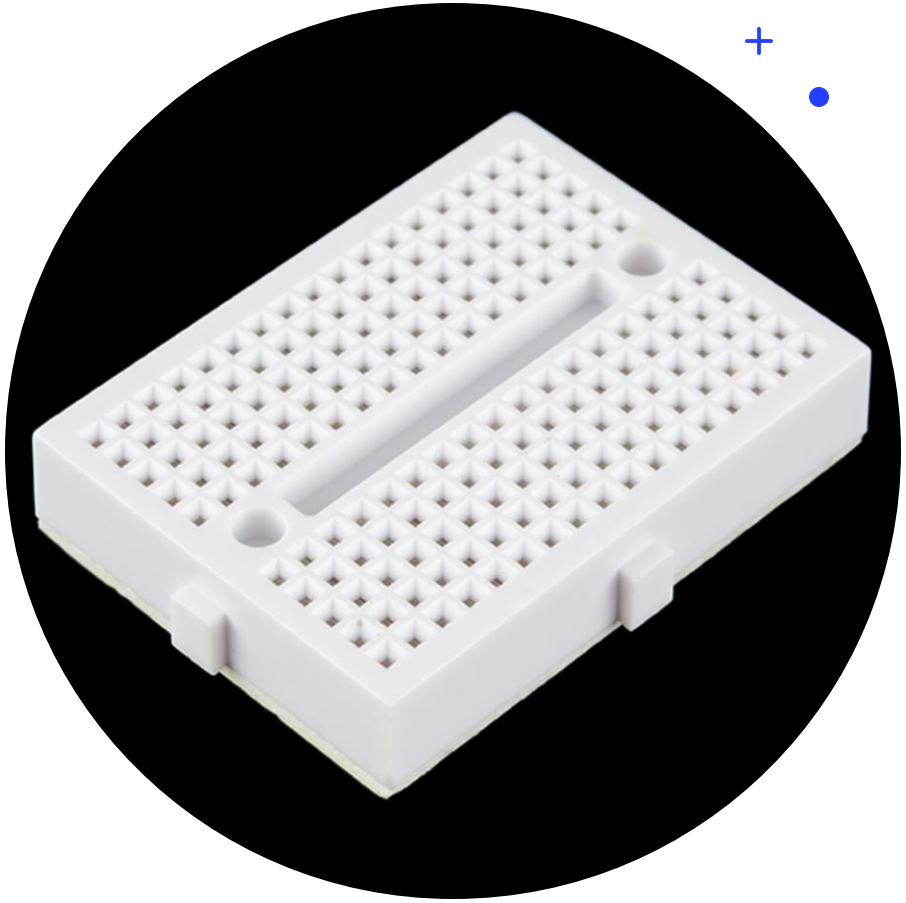
Component 6: 3x9v soldered Battery

- The 3x9v parallel connected dry cells provide power to the entire forklift system.
- It supplies 9v of energy to the Arduino Uno, actuators and sensors.
- Using a portable battery allows the forklift to operate wirelessly and move around freely
- Allows for continuous operation due to our project requiring high power



Component 7: Breadboard

- Due to the large number of components, the Arduino Uno did not have enough pins to power everything directly.
- A breadboard is used to share power and ground across all components, also initially for the servo motor 1 and 2 to share the same signal, however we have scrapped servo motor 2
- This creates a common power supply and common ground, ensuring stable tidy operation and easy troubleshooting



COMPONENT 8: LCD

- The LCD provides real-time feedback by displaying system states, user prompts, and safety alerts generated directly by the program.

- The next slide will explain about some of these parts, which will indicate different states

COMPONENT 8: LCD

- The LCD updates the cars current status like:
- “Touch to Start” – prompts user interaction
- “Target Set: 3” – confirms selected destination
- “Heading 3” – shows navigation progress
- “OBSTACLE! 18cm” – alerts when an object is detected
- “Action: Pickup Sequence” – indicates task execution
- “Returning” – shows return-to-home mode
- “EMERGENCY LOCKED” – safety lock activation
- “Mission Done” – confirms task completion

Component 9: Buzzer

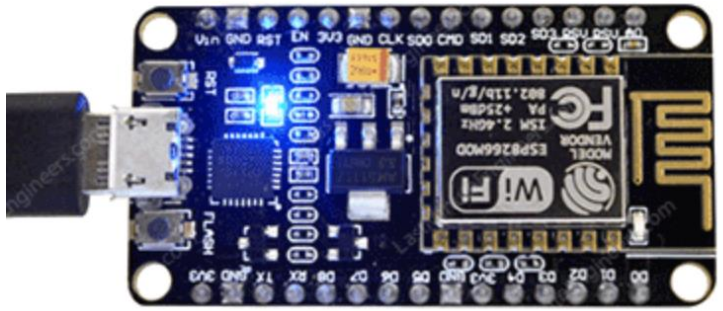
- The buzzer is used to signify changes in the forklift's state to give auditory feedback during forklift ops

COMPONENT 10: MOTORS

Used to physically move the forklift and motor while working with the ir sensors on direction throughout the simulated warehouse

Component 11: Wifi module

- Allows wireless operation, monitoring, and configuration of the forklift
- **M** → Status Report: Shows live sensor readings, motor config, battery voltage, and past trips (EEPROM)
- **+ / –** → Motor Speed: Adjust speed and save to EEPROM
- **S** → Emergency Stop / C → Admin Unlock: Stop and unlock forklift remotely
- **R** → Reset System
- **H** → Go Home
- **IP** → Wi-Fi IP Request
- **1 / 2 / 3 / 4** → Set Destination
-



How does our solution support long term sustainability and resilience

Designed for long-term wellbeing, not just convenience
Reduces human exposure to dangerous forklift operations, improving worker safety and health over time.

Lowers operational risks
Sensor-based obstacle detection minimizes collisions, accidents, and equipment damage in warehouses.

Adapts to future changes
Modular Arduino-based design allows easy upgrades with new sensors, IoT connectivity, or AI features as warehouse needs evolve.

Improves resource efficiency
Optimized movement and automation reduce wasted time, energy, and operational errors.

Minimizes negative environmental impact
Efficient autonomous transport lowers unnecessary vehicle movement and supports energy-conscious warehouse operations.

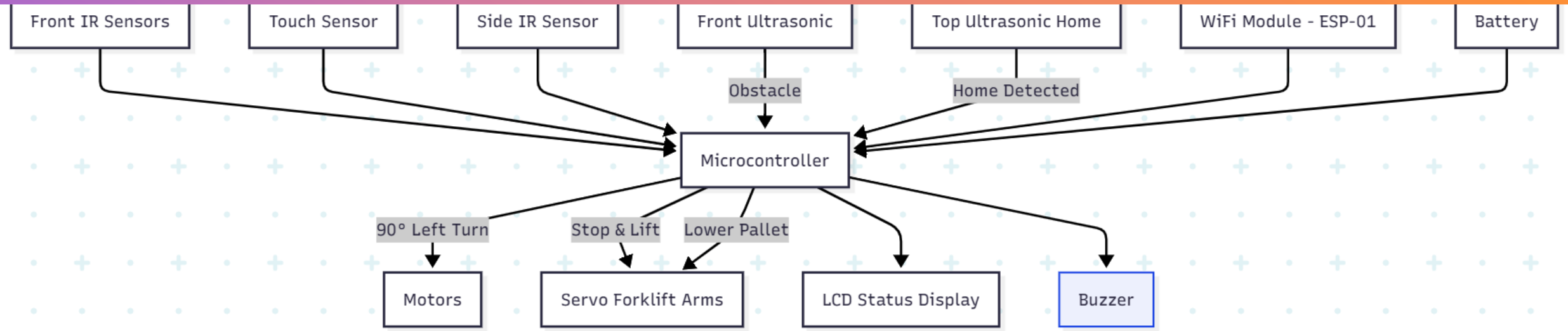
What Trade-offs does our design have?

Improved safety vs. fewer manual jobs
Automating forklift work makes warehouses safer, but it may reduce the need for manual operators. I believe this should be balanced by helping workers move into roles like system monitoring, maintenance, or technical support.

Affordable design vs. limited performance
Using simple hardware keeps costs low, but the system cannot handle very complex tasks like industrial forklifts.

Efficiency vs. human judgment
The robot works well for routine transport, but humans are still better at handling unexpected situations.

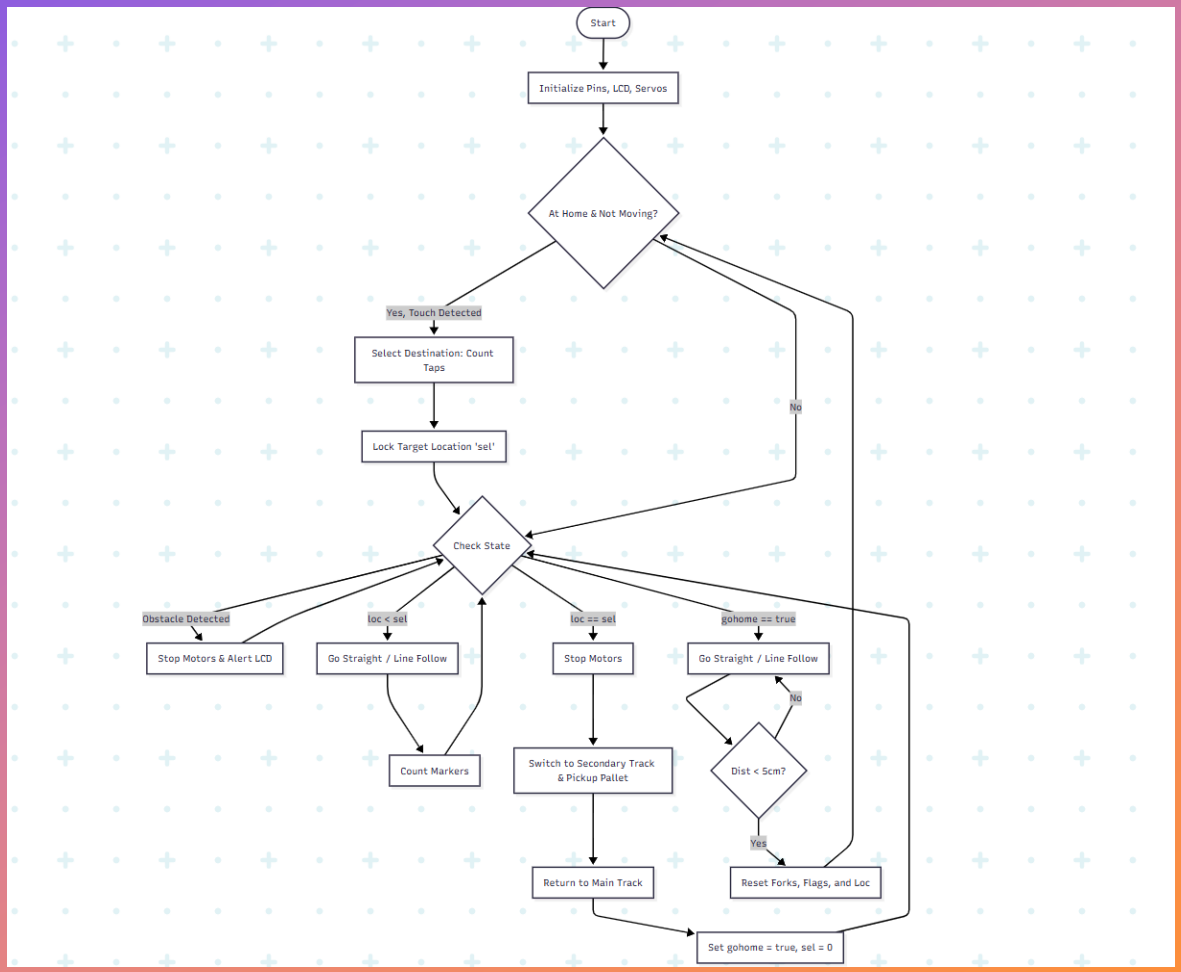
Dependence on sensors vs. reliability
Sensors help prevent accidents, but failures or inaccurate readings could affect operation, so regular maintenance is important.

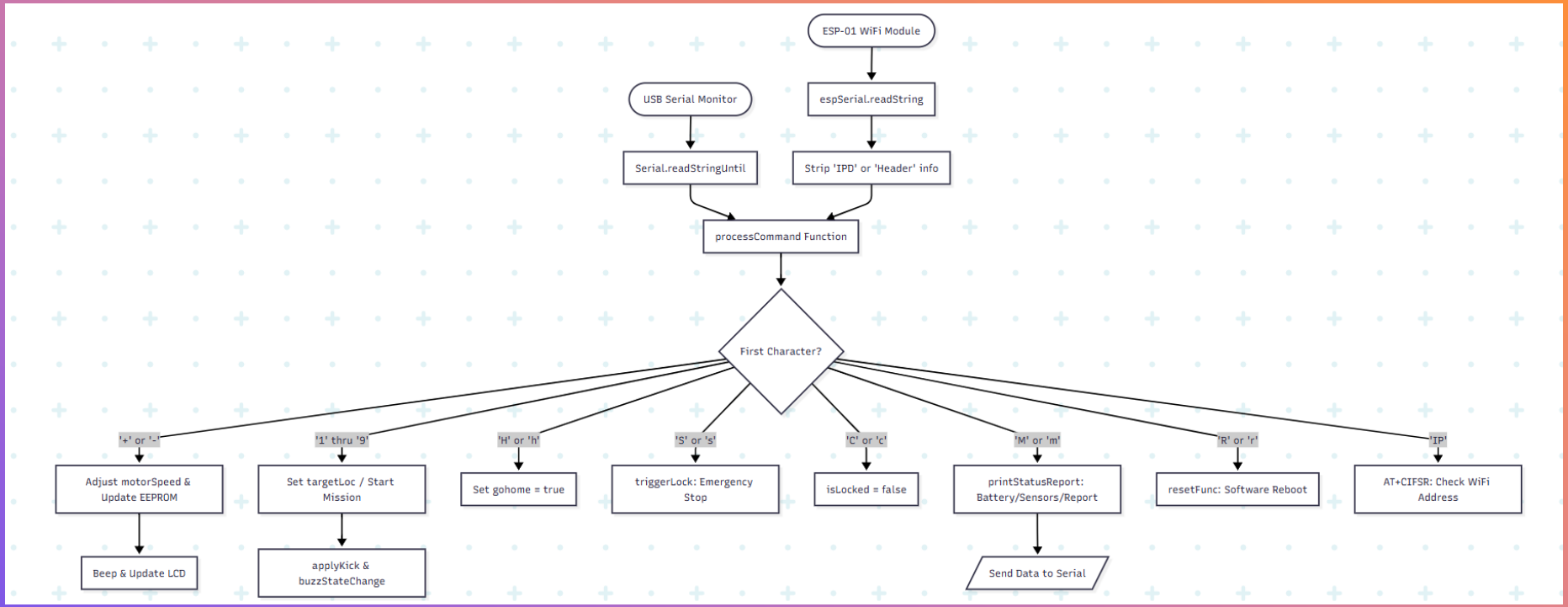


BLOCK DIAGRAM

FLOW CHART

Flow Chart explaining the main process

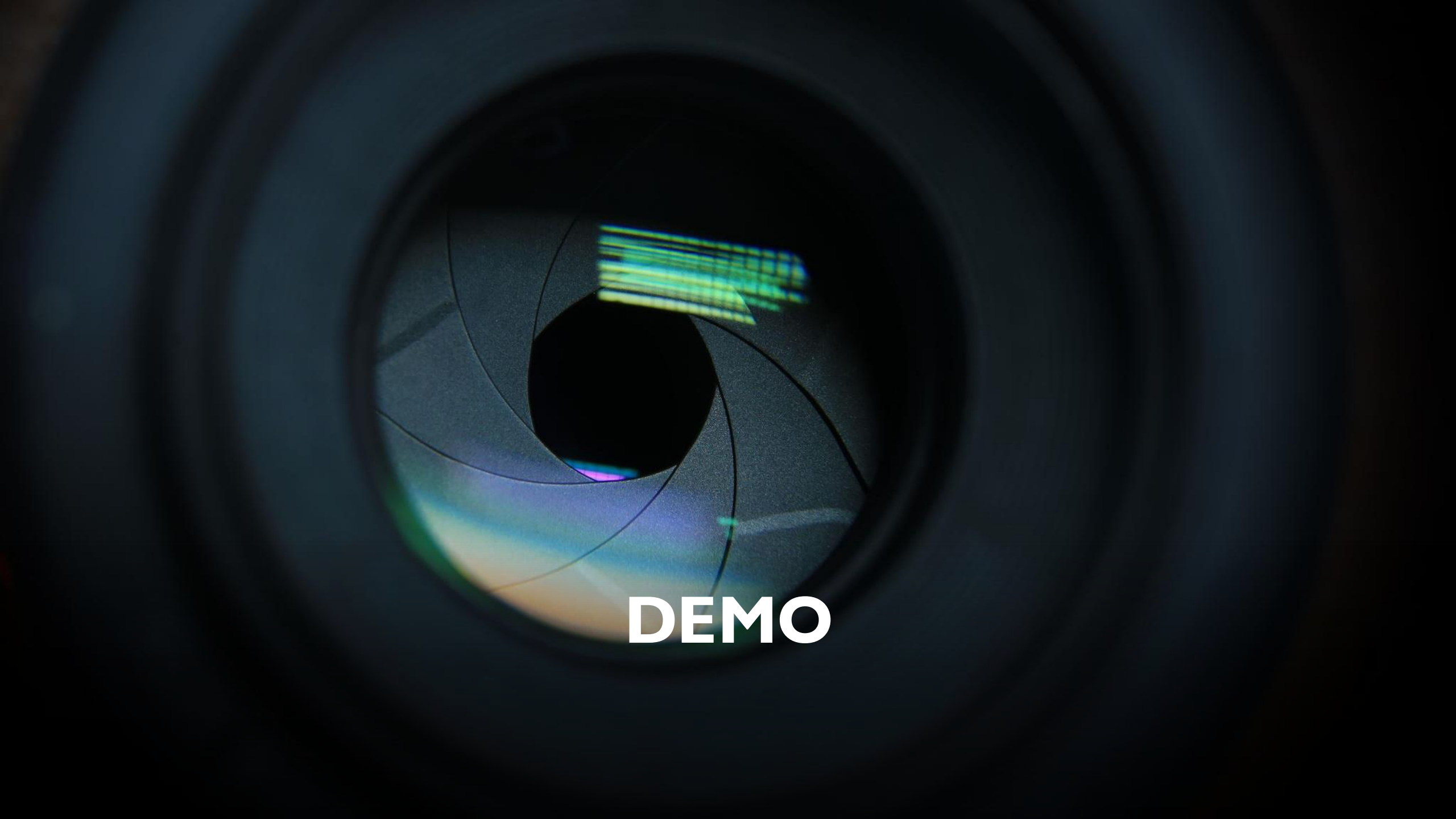




FLOW CHART

Flow chart explaining the process with Serial input

Project Video



DEMO

Code

- Without Wi-Fi(for faster bootup)

forklift_v13.ino

- With Wi-Fi(full code)

forklift_v14.ino

Contributions (wynn)

- Wynn: helped with debugging, modifying overall code,helped w refining executepickuproutine() to ensure that the car aligns with the track
- Problem solving and troubleshooting for physical hardware/connection issues(motor ground wire coming off, battery pack vcc coming off) & directed overall car structure decision making, testing original sample codes for motor,lcd,ir sensor for finding out issue which was insufficient current to arduino
- Used solid edge to build entirety of car body as well as necessary sensor housings, directed how the car would stay intact, however due to lack of time we did not use the wire to tie the car to the body using the tiny holes,
- In charge of final 3d printing and drafting (3 past 3d prints), added on to the car model to integrate different components, in charge of securing components and wiring to 3d car like servo motor (+printing servo holder), touch sensor secured via wire, ultrasound secured inside using wires as well, front ultrasound secured via wire tying, securing lcd via plastic screws , in charge of building and putting together lego for the forklift, in charge of structural decision making
- Slightly modified slides to better fit code (eeprom and wifi setup)

Contributions (Skyler)

- Skyler: debugging, modifying, refining Zhili's code (V11 and V12)
- In-charge of the process of turning left after protrusion IR counts black tape. Timed the **(Corrected_Turn_Left function)** to be around 90 degrees, black line following **(drive_logic function)**, changed **(updateLCDStatus function)** removed lcd.clear() to prevent LCD from flickering when driving.
- Worked on the flow of the **(executePickupRoutine function)** make sure that the car does pickup in this order, turn left --> stop for a while --> align with guiding line --> when sense object servo will lift --> drive home.
- Added **(forkliftup and forkliftdown function)** so servo motor will move by 10 degrees each count for 9 times, from degrees to 90 degree or the other way round.
- Added ignore first marker condition in (**count_markers function)** because there was a problem when selecting loc 1, it will immediately do corrected_turn_left function.
- Video forklift performing drive_logic and corrected_turn_left.
- In charge of forklift lego piece building
- Soldering of a battery pack, did sketch of the black line track, design and building of the black line board
- In charge of 3d printing and blueprinting the initial best fit car model, helped with measurements of components for Wynn to draft out, also helped with securing car and alot of ideation

Contributions (Justin)

- Justin: worked on slides, helped with parts of sketch, helped with design and building of black line board
- Did functional block diagram and flowchart
- Also helped with building parts of the forklift with Wynn
- Helped abit with the construction of the battery component with skyler

Contributions (zhili)

- Zhi li: did 85-90% of the code and functions and explained it to all group members, as well as researching on how to implement new (not taught) components; and integrating each versions made by wynn and skyler with the newer components, designed &directed overall code and system flow, helped justin with functional block diagram and flowchart, helped with cable management

References

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